FIITJEE

ALL INDIA TEST SERIES

CONCEPT RECAPITULATION TEST – II

JEE (Advanced)-2021

PAPER – 2

TEST DATE: 13-06-2021

ANSWERS, HINTS & SOLUTIONS

Physics

PART – I

SECTION – A

1. ABC

2. AD

3. ACD

4. AD

5. ACD

6. ACD

SECTION – B

7. 6

8. 1

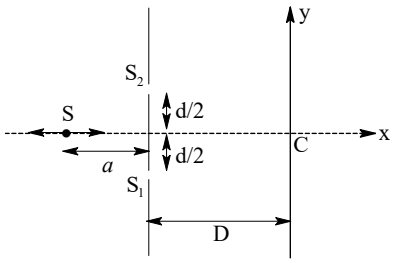
Sol. In the CM frame, work done by pseudo forces $= 2\left(\frac{F}{2}\times\ell\right) = F\ell$, this is the energy lost in collision.

9. 6

Sol. $I = \dfrac{P}{4\pi r^2}$

$$I = \frac{P}{4\pi\left(a^2 + \left(\frac{d}{2}\right)^2\right)} \Rightarrow I_{max} = 4I$$

Differentiate to find answer



10. 3

11. 3

12. 3

SECTION – C

13. 00001.20

14. 00000.50

15. 00000.40

16. 00002.99

17. 00001.67

18. 00001.00

Sol. The change in internal energy of the gas is DU = 1.5 nR (DT) = 1.5 (1 mol) (253J/mol-K) (2K) = 25 J.
The heat given to the gas = 42 J
The work done by the gas is DW = DQ – DU= 42 J – 25 J = 17 J.
If the distance moved by the piston is x, the work done is DW = (100 kPa) (8.5 cm2) x.
Thus, (105 N/m2) (8.5 × 10–4 m2) x = 17 J or, x = 0.2m = 20 cm.

Chemistry

PART – II

SECTION – A

19. ABCD

20. CD

21. ABD

22. ABC

23. ACD
Sol. B is wrong so $\rightarrow$ACD are correct

24. ACD
Sol. Mo in Haber's process is promoter catalyst is iron oxide with K_2O and Al_2O_3

SECTION – B

25. 5

26. 4

27. 6

28. 5

29. 2
Sol. $\lambda_{CH_4} = \frac{h}{m_{CH_4} \times v_{CH_4}}, \lambda_{O_2} = \frac{h}{m_{O_2} \times v_{O_2}}$

$\frac{v_{CH_4}}{v_{O_2}} = \frac{m_{O_2}}{m_{CH_4}} = \frac{32}{16} = 2$

30. 4

SECTION – C

31. 00209.72
Sol. Given $\Delta E = +0.21 kJ\,mol^{-1} = 0.21 \times 10^3\,J\,mol^{-1}$

$P = \bar{1} = 1.0 \times 10^5\,Pa$

$= \left(\frac{100}{2.93} - \frac{100}{2.71}\right) cm^3\,mol^{-1}$ of $CaCO_3$

$= -2.77\,cm^3 = -2.77 \times 10^{-6}\,m^3$

$\therefore \Delta H = 0.21 \times 10^3 - 1 \times 10^5 \times 2.77 \times 10^{-6} = 209.72$

32. 00000.05

Sol. $\left[Ag(NH_3)_2\right]^+(aq) \rightleftharpoons Ag^+(aq) + 2NH_3(aq)$

$x+y \qquad 2x$

$AgCl(s) \rightleftharpoons Ag^+(aq) + Cl^-(aq)$

$x+y \qquad y$

In case of simultaneous solubility, Ag^+ remains same in both the equilibrium

$$K_c = \frac{(x+y)\times 2^2}{\left[Ag(NH_3)_2\right]^+} \quad .....(1)$$

$$K_{sp} = (x+y)\times y \quad .....(2)$$

$$\therefore \frac{K_c}{K_{sp}} = \frac{(2x)^2}{\left[Ag(NH_3)_2\right]^+ \times y} \text{ Given, } [NH_3] = 2x = 1M$$

$\left[Ag(NH_3)_2\right]^+ = [Cl^-] = y$ because Ag^+ obtained from $AgCl$ passes in $\left[Ag(NH_3)_2\right]^+$ state.

$$\frac{K_c}{K_{sp}} = \frac{1}{y\times y} \Rightarrow y^2 = \frac{2.378\times 10^{-10}}{8.2\times 10^{-8}} = 0.29\times 10^{-2} \therefore y = 0.539\times 10^{-1} = 0.0539M$$

That is, $\left[Ag(NH_3)_2\right]^+ = 0.0539M$

33. 00000.44

Sol. $N_2 + 3H_2 \rightleftharpoons 2NH_3$

$\Delta n = -2$ and $K_p = K_c(RT)^{\Delta n}$

$3.28 = K_c(0.0821\times 673)^{-2}$ and $K_c = 1.0\,\text{mole}^{-2}\text{litre}^2$

34. 00006.67

Sol. $BaCl_2 + Na_2SO_4 \rightarrow BaSO_4 + 2\,NaCl$

0.04 moles 0.05 0 0

0 0.01 Not effect 0.08

$\frac{0.01}{0.7}$ Not effect $\frac{0.08}{0.7}$

$$= \left(3\times\frac{0.01}{0.7} + 2\times\frac{0.08}{0.7}\right)0.082\times 300 = 6.67\,\text{atm}$$

35. 00005.51

Sol. The overall reaction is:

$Pb_{(S)} + PbO_{(2s)} + 2H_2SO_4 \longrightarrow 2PbSO_{4(s)} + 2H_2O$

$\therefore$ Moles of electron used $= \frac{3.5\times 6\times 60\times 60}{96500} = 0.783$

$\therefore$ Molarity of H_2SO_4 left $= \frac{8.817}{1.6} = 5.51$

36. 00000.33

Sol. Let $x = \text{moles of } C_2H_6 \text{ and } y = \text{mols of } C_2H_4$

Moles of O_2 used $= \left(\frac{7x}{2} + 3y\right)$

Moles of O_2 available = 130/32

$$\therefore \frac{7x}{2} + 3y = \frac{130}{32}$$

$$\Rightarrow 3.5x + 3y = 4.06$$

Also, $x + y = \frac{PV}{RT} = \frac{1 \times 40}{0.0821 \times 400} = 1.218$

Solving for x and y, we get x = 0.82 and y = 0.398

$$\therefore x_{C_2H_6} = \frac{0.812}{1.218} = 0.67 \text{ and } y_{C_2H_4} = 0.33$$

Mathematics

PART – III

SECTION – A

37. C, D

Sol. $a+b+\sqrt{a^2+b^2}=\frac{ab}{2}\Rightarrow b=\frac{4(a-2)}{a-4}$

$\Rightarrow a=5,b=12,a=6,b=8$.

38. A, D

Sol. $\sqrt{x^2+y^2+2y+1}=\sqrt{x^2+(y+1)^2}$ is the distance between point (x,y) and $(0,-1)$. The distance from (x,y) to the line $x-2y+3=0$ is $\frac{|x-2y+3|}{\sqrt{5}}$

The ratio of these 2 distances is the constant $\frac{\sqrt{5}}{\sqrt{\lambda}}$ for ellipse $\sqrt{\frac{5}{\lambda}}<1\Rightarrow\lambda>5$.

39. B, C

Sol. $a_{n/2}$ and $a_{(n+1)/2}$ are both 1, Note that the number of ways to choose numbers left of $a_{n/2}$ and $a_{\frac{n+1}{2}}\left(a_1>a_2>a_3...a_{\frac{n}{2}};a_{\frac{n+1}{2}}\right)$ is same as that of right. We started with 9

$n=9\Rightarrow {}^8C_4=70(\text{rejected})$

$n=10\Rightarrow {}^9C_4=126(\text{rejected})$

$n=11\Rightarrow {}^{10}C_5=252$

$n=12\Rightarrow {}^{11}C_5=462$

$n=13\Rightarrow {}^{12}C_6=924(\text{rejected})$.

40. A, C, D

Sol. $\lambda^3 = -a\lambda^2 - b\lambda - c$

$\Rightarrow \lambda^4 = -a\lambda^3 - b\lambda^2 - c\lambda = (1-a)\lambda^3 + (a-b)\lambda^2 + (b-c)\lambda + c$

if $|\lambda|\geq 1$, then $|\lambda|^4\leq(1-a)|\lambda|^3+(a-b)|\lambda|^2(b-c)|\lambda|+c$

$\leq (1-a)|\lambda|^3+(a-b)|\lambda|^3+(b-c)|\lambda|^3+c|\lambda|^3\leq|\lambda|^3$

$\Rightarrow |\lambda|\leq 1$

Hence the only possibility is $|\lambda|=1$

Then $|\lambda|\leq 1$.

41. A, C

Sol. $A^3+B^3+C^3=A^3+B^3+C^3-3ABC$

$=(A+B+C)(A^2+B^2+C^2-AB-BC-CA)$

$=(A+B+C)(A+WB+W^2C)(A+W^2B+WC)$ [W = cube root of unity]

$=(A+B+C)(A+WB+W^2C)\overline{(A+WB+W^2C)}$

thus $\det(A^3+B^3+C^3).\det(A+B+C)$

$\det(A + B + C)^2.\det(A + WB + W^2C)\det\overline{\left(A + WB + W^2C\right)}$

$= \left(\det\left(A + B + C\right)\right)^2 . \left|\det\left(A + WB + W^2C\right)\right|^2 \geq 0$.

42. A, C

Sol. Put x = 1990 in(i) we get $\left[\frac{f(1990)}{90}\right]$ = f(1990) – 14 …….(0)(iii)

Using relation(ii) $\frac{1990}{90} < \frac{f(1990)}{90} < \frac{2000}{90}$

$\Rightarrow \left[\frac{f(1990)}{90}\right]$ = 21 or 22

of $\left[\frac{f(1990)}{90}\right]$ = 21 $\Rightarrow$ f(1990) = 1904 Using (iii)

of $\left[\frac{f(1990)}{90}\right]$ = 22 $\Rightarrow$ f(1990) = 1994 using(iii)

SECTION – B

43. 1

Sol. $x^2 + 2ax – 16 = 0$ is the quadratic equation whose roots are abscissa of A, B.
$y^2 + 24y – 9 = 0$ is the quadratic equation whose roots are ordinates of A, B.
Equation of circle with AB as diameter is $(x^2 + 2ax – 16) + (y^2 + 24y – 9) = 0$

Radius = $\sqrt{a^2 + 12^2 + 16 + 9}$

$\Rightarrow R^2 = a^2 + 13^2 \Rightarrow (R – a)(R + a) = 169.1 \Rightarrow R – a = 1$

44. 2

Sol. Any point on the line is $\left(\frac{r}{2}, \frac{\sqrt{3}r}{2}\right)$, where r is the distance from the origin.

$\frac{r^3}{8} + \frac{3\sqrt{3}r^3}{8} = \frac{\sqrt{3}}{4}r^2 + 2$

$\Rightarrow \frac{\left(1 + 3\sqrt{3}\right)}{8}r^3 - \frac{\sqrt{3}}{4}r^2 - 2 = 0$

$r_1r_2r_3 = \frac{2 \times 8}{1 + 3\sqrt{3}} = \frac{16}{1 + 3\sqrt{3}}$

$[r_1r_2r_3] = 2$

45. 5

46. 1

Sol. Let Radius of new circle = r
From symmetry centre of the required circle is (0, 0)

$\Rightarrow C_1O = \sqrt{2\left(\sqrt{2} - 1\right)^2} = 2 – \sqrt{2} = r – (\sqrt{2} – 1)$

$\Rightarrow r = 1$

C_1O is the distance from origin to centre of any of four circles

47. 3

Sol. $\because \dfrac{a+\frac{b}{2}+\frac{b}{2}+\frac{c}{3}+\frac{c}{3}+\frac{c}{3}}{6} \geq \sqrt[6]{a\cdot\frac{b}{2}\cdot\frac{b}{2}\cdot\frac{c}{3}\cdot\frac{c}{3}\cdot\frac{c}{3}}$

48. 5

Sol. If $x^3 + ax + 1 = 0$ has a repeated root, then $3x^2 + a = 0 \Rightarrow x = \pm\sqrt{\dfrac{-a}{3}}$ should satisfy

$x^3 + ax + 1 = 0$

$\therefore \tan\theta = -\left(\dfrac{27}{4}\right)^{1/3}$.

SECTION – C

49. 00002.00

50. 00009.00

Sol. r_1, r_2, r_3 are the roots of the cubic

$x^3 - (r + 4R)x^2 + s^2x - s^2r = 0$.

51. 00005.00

Sol. Number of zeroes = $x = \sum_{i=1}^{\infty}\left[\dfrac{n}{5i}\right]$

$x < \sum_{i=1}^{\infty}\left[\dfrac{n}{5i}\right]$ since $\left[\dfrac{n}{5i}\right] < \dfrac{n}{5i}$

$x < \dfrac{n}{5}\sum_{i=1}^{\infty}\dfrac{1}{5^{i-1}}$

$\Rightarrow x < \dfrac{n}{5}\times\dfrac{5}{4}$

$\Rightarrow n > 4x \Rightarrow n > 80$

If $n = 80$ $x = \left[\dfrac{80}{5}\right] + \left[\dfrac{80}{25}\right] = 16 + 3 = 19$

$n = 85$ $x = \left[\dfrac{85}{5}\right] + \left[\dfrac{85}{25}\right] = 20$

$n = 90$ $x = \left[\dfrac{90}{5}\right] + \left[\dfrac{90}{25}\right] = 21$

$\Rightarrow n \in \{85, 86, 87, 88, 89\}$

52. 00004.00

Sol. $x^2dy + xydx = y^3(xdy - ydx)$

$\Rightarrow xd(xy) = y^3x^2d(y/x)$

$\Rightarrow \dfrac{d(xy)}{x^2y^2} = \dfrac{yd(\frac{y}{x})}{x}$

$\Rightarrow \dfrac{y^2}{2x^2} + \dfrac{1}{xy} = C$ (C is an integration constant)

Put $x = 1 \Rightarrow C = \dfrac{y^3 + 2}{2y} = 0$

$\Rightarrow y^3 = -2x$ is the function.

$\Rightarrow -8 = -2x \Rightarrow x = 4$

53. 00002.00

54. 00008.00

Sol. ΔBMH and ΔCMT are similar, $\angle MBH = \angle MCT$,

$\Rightarrow BH \parallel CT$, $CT \perp AC$ and $CT = BH$

Now, $AT^2 = TC^2 + AC^2 = BH^2 + AC^2$

$= BD^2\text{cosec}^2C + AD^2\text{cosec}^2C$

$= \dfrac{AB^2}{\sin^2 C} = \dfrac{a^2}{\sin^2 A} = 16 \times 4 = 64 \Rightarrow AT = 8.$

